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Data and Robustness Modeling

APS

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CHECKPOINT

ADAPTIVE PATHOGEN SURVEILLANCE | INDIVIDUAL PROGRESS BRIEF

MY ROLE

I work on turning simulated outbreaks into organized datasets that our models can learn from and be tested against. I also examine how noise and missing sensor readings change the difficulty of finding a source.

OUR WORKING FOUNDATION

Our simulator can create sensor observations while keeping track of the true source for each run. I am using this foundation to organize examples across different source zones and sensor conditions. These are synthetic examples, not real patient or laboratory data.

I want each experiment to be reproducible, meaning another teammate can regenerate the same examples from the saved settings and random seed.

MY CHECKPOINT STATUS

I am: [building / testing / refining]

MY CONTRIBUTION SO FAR

I have completed: [specific file, test, or output]

TERMS AND IDEAS I USE

A synthetic dataset is a collection of computer-generated examples. Each run is one simulated outbreak; its features are the sensor readings given to a model, and its label is the true source zone used to check the answer. Noise means a reading varies from its expected value. Missing data means no reading is available. A training set teaches a model patterns; a separate test set measures how it performs on examples it has not seen.

WHAT I AM WORKING ON NEXT

1. I will agree with Emmanuel and Jimmy on consistent columns for run ID, source label, sensor readings, missing values, and simulation settings.
2. I will generate examples from different source zones while varying noise and missing readings in a controlled way.
3. I will document the dataset and keep independent training and test runs separate so our results remain fair.

CHALLENGES I AM CONSIDERING

My main concern is data leakage, which happens when test examples are too closely related to data the model already learned from. I also need a consistent way to represent missing sensors.

WHAT I PLAN TO SHOW

I plan to show a sample dataset, explain what one row represents, and identify the settings used to generate it.

HOW I WOULD EXPLAIN MY WORK

My checkpoint summary: My part is to provide reliable synthetic data for the project. I know the simulated source of each run, so I can label the examples and help the team measure whether its predictions are correct. I am focusing on consistent data, realistic testing conditions, and repeatable experiments.